

A standardized framework for circulating blood proteomics

Received: 3 March 2025

Accepted: 29 July 2025

Published online: 23 September 2025



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The circulating blood proteome holds immense potential for biomarker discovery and understanding disease mechanisms. Notable advances in mass spectrometry and affinity-based technologies have been made, but data integration across studies and platforms is hindered by the absence of unified analytical standards. This limitation impedes comprehensive exploration of human biology across diverse phenotypes and cohorts as well as the translation of findings into clinical applications. The disparities between datasets, stemming from a combination of factors related to differences in sample collection, pre-analytical handling, measurement methods and instrumentation, further complicate data integration. In this Perspective, we outline key challenges in blood-based proteomics and propose actionable strategies. Central to our recommendations are high-quality, technology-agnostic reference samples, which can bridge disparate datasets and enable robust cross-study comparisons. By fostering interconnected investigations across proteomic technologies, blood sample collections, clinical phenotypes and different populations, these references will accelerate the field and its translation.

The circulating blood proteome is an invaluable resource for studying human health due to minimally invasive access and clinically validated biomarkers, which dynamically reflect physiological and pathological states^{1,2}. Current proteomic studies focus mostly on comparing blood protein composition and dynamics, providing critical insights for biomedical research and human phenotypes¹. Among the most impactful applications, the UK Biobank³ reported protein profiles in plasma samples from over 50,000 individuals^{4–6}, complemented by extensive phenotypic and genetic data. These plasma proteome data are based on one standardized protocol, primarily the proximity extension assay (PEA)⁷, allowing researchers worldwide to reanalyze data without concerns about analytical methods or sampling differences. Until now, only scientists conducting PEA-based studies could directly compare their data with the UK Biobank dataset. Several affinity proteomic

platforms exist, including assays built on slow off-rate modified aptamers (SOMAmers)⁸, the nucleic acid-linked immuno-sandwich assay (NULISA)⁹ and Meso Scale Discovery (MSD)¹⁰, each based on distinctive proprietary reagents. The PEA, NULISA and MSD platforms use pairs of highly specific, high-affinity antibodies for sensitive detection of low-abundance analytes, using DNA sequencing, quantitative PCR or electroluminescence as the readout. SOMAmers use modified DNA aptamers as scaffolds with slow off-rates to capture analytes and serve as readout elements. While studies show high comparability within each platform, considerable differences exist between these analytical approaches in protein content and performance.

While affinity-based proteomic methods use ready-made kits with excellent batch consistency, their scope remains limited to pre-selected protein panels covering <50% of the human proteome.

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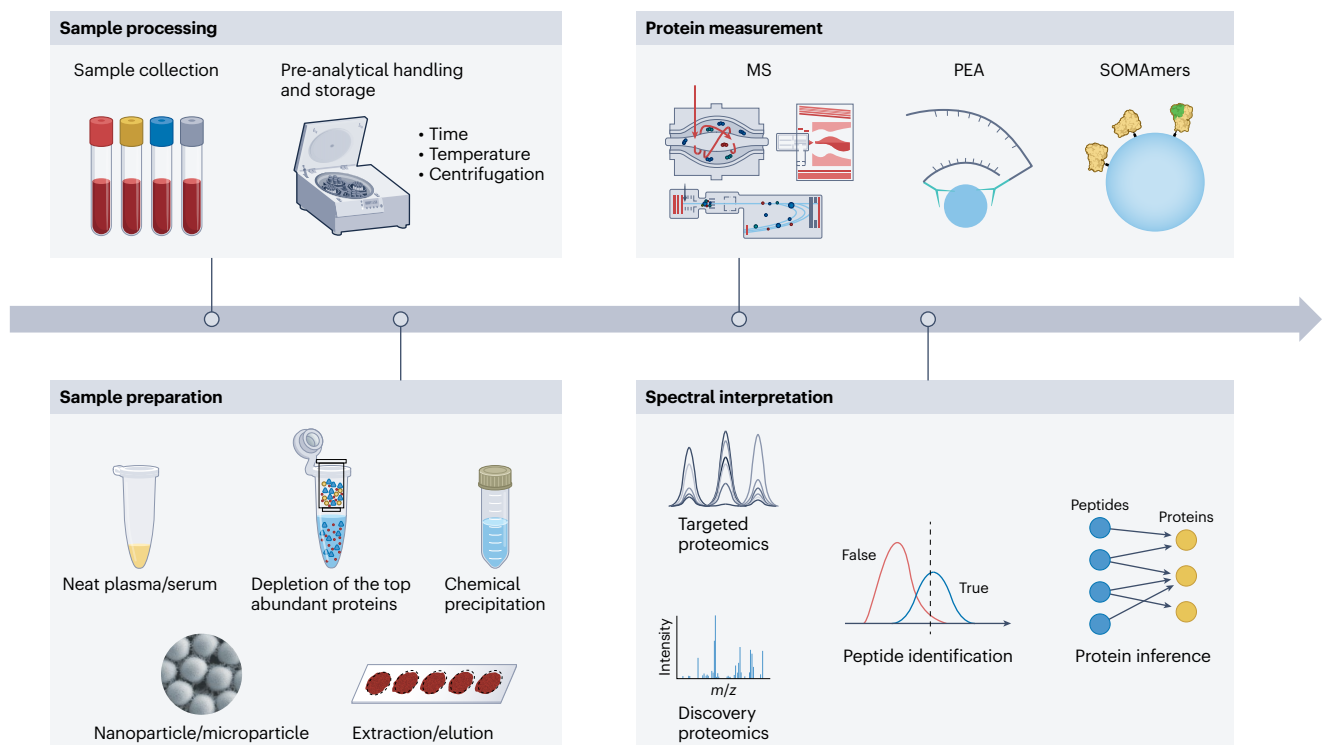


Fig. 1 | Workflows for analyzing the circulating blood proteome. Analyzing the circulating blood proteome involves multiple steps, including sample collection, pre-analytical handling and storage, preparation methods, protein measurement techniques and spectral interpretation. Protocols for each step vary widely.

Notably, sample preparation (except for neat plasma processing) and spectral interpretation are essential components of MS-based proteomic workflows but do not apply to affinity-based techniques, which use direct binding assays. Figure created using BioRender. lab, g. (2025) <https://BioRender.com/1w193ye>.

Although Antibodypedia is a catalog of millions of antibody products targeting over 19,000 human gene products (covering approximately 95% of all human genes), this broad coverage does not guarantee the performance required for sensitive detection¹¹. Moreover, proteoforms¹², which have crucial roles in fundamental biological processes and represent the complex interplay between human genetics and diverse health and disease phenotypes¹³, expand the analytical scope of the circulating blood proteome beyond canonical sequences. Historically, most antibodies have not been produced or investigated to differentiate between these variations. Owing to the bias favoring certain (popular) proteins for antibody generation, gaps remain, particularly for understudied proteins^{14,15} and most protein isoforms that create functional diversity. For instance, PTEN α , an isoform of the tumor suppressor PTEN, has been reported to regulate metabolic functions¹⁶, but the current lack of isoform-specific antibodies necessitates reliance on antibodies that measure total PTEN protein levels¹⁷. By contrast, proteomics based on mass spectrometry (MS) can identify both known and unknown proteins as well as their proteoforms, such as splicing variants, post-translational modifications (PTMs) and other sequence variations. Still, MS-based protein measurements are biased toward higher-abundance proteins¹⁸. Because of the high dynamic range of protein concentrations, sample preparation techniques such as depleting abundant proteins or enriching low-abundance proteins are essential for obtaining balanced protein profiles with enhanced sensitivity.

This Perspective highlights the critical need for standardized blood proteome data generation. We discuss the urgency of establishing reliable reference materials and present practical solutions.

The complexity of MS-based circulating blood proteomics

Since the launch of the Human Plasma Proteome Project by the Human Proteome Organization (HUPO) in 2002 (ref. 19), hundreds of plasma

proteome datasets^{20–22} have been generated by the community using MS. However, cross-study comparisons remained challenging due to the complexity and continuous development of different proteomic workflows and the lack of integration of a unified standard sample in different studies²³.

These workflows include sample collection and processing, pre-analytical handling, storage, sample preparation, protein measurement and spectral interpretation, with each step subject to optimization, causing substantial variations (Fig. 1). Additionally, multiple statistical analysis options are available, tailored for different purposes, such as biomarker discovery, prognostic prediction and disease monitoring.

Bias risk begins with sample collection, in which various collection tubes separate liquid from blood cells before biochemical analyses. Aside from serum, EDTA plasma is the most widely collected and used². Pre-analytical handling, including needle-to-freezer time variations, centrifugation techniques, temperatures, erythrocyte lysis or contamination²⁴, and platelet activation or contamination²⁵, profoundly impact the proteome. While MS-based proteomics is relatively insensitive to peptide-level variations from protein unfolding during sample storage and transportation²⁶, it is influenced by protein degradation, as environmental temperatures and freeze–thaw cycles introduce extensive variations between collection sites and sample sets.

Different methods can then be used in sample preparation for MS analysis, including the use of neat plasma, nanoparticle- or microparticle-based low-abundance protein enrichment^{27–29}, depletion of the top abundant proteins^{30,31}, chemical precipitation by perchloric acid³² or fractionation³³. These methods are selected based on specific analytical goals to increase the possibility of detecting more proteins per sample, hence offering deeper proteome coverage due to higher sensitivity but aiming to maintain the reproducibility of the MS-based results.

MS instruments have been continuously improved to enhance accuracy, sensitivity, resolution and scanning rate and are operated in either untargeted or targeted modes¹⁸. In the untargeted mode, data-dependent acquisition (DDA) selects the most-intense precursor ions for fragmentation, while data-independent acquisition (DIA) systematically fragments all ions within predefined mass windows. Both approaches are widely used in clinical proteomics to maximize proteome coverage and identify potential biomarkers. Additionally, DDA is often combined with labeling techniques, such as tandem mass tagging³⁴ and isobaric tags for relative and absolute quantitation³⁵, enabling high sample multiplexing, reducing liquid chromatography (LC)–MS/MS runs and enhancing protein quantification accuracy. Fractionation methods, such as high-pH peptide separation before MS detection, are also used to improve proteome coverage and enable deeper protein identification³⁶. In the targeted mode, selected or multiple-reaction monitoring³⁷ and parallel reaction monitoring^{38,39} are the most commonly used acquisition modes, focusing on proteins of interest. Most often used with internal references, these targeted MS methods allow for consistent and precise quantification over time.

MS-based proteomics requires advanced software solutions for analyzing data obtained from instruments provided by different companies. Analysis of the same blood proteome data using different software solutions can yield slight variations in results. Comparing different DIA analysis software yields generally comparable results with some noticeable variations^{40,41}. Additionally, parameter settings during data processing can also contribute to variations in protein identification and quantification. This includes discrepancies in FASTA files (whether the FASTA files consist of reviewed or unreviewed proteins from sources like UniProt and NCBI) and match-between-runs, cross-run normalization, protein inference and false discovery rate.

Variability in proteomic studies arises not only from technical platforms but also from cohort characteristics, sample handling and batch effects. Certain proteins may only be detectable when elevated in specific disease states. Additionally, proteoforms with specific PTMs, single-nucleotide polymorphisms or other sequence variants may only become detectable under certain physiological or pathological conditions, potentially causing variations in protein measurements using affinity-based tests and MS. Furthermore, some proteins are prone to degradation over time in cohort samples that have been stored for decades. These factors should not be overlooked and can be effectively evaluated using appropriate reference materials.

These complexities lead to extensive diversification of MS-based datasets from the circulating blood proteome that is independent of biological or phenotypic variability. If this diversification persists, much potentially insightful data may remain unsuitable for comparative studies, hence, not being sustainable. This challenge arises from the wide range of possible samples, phenotypes, populations and workflow variations. We currently lack adequate tools to effectively manage, standardize and resolve these differences.

Investigating method-specific differences between circulating proteomics datasets

Affinity-based methods like PEA and SOMAms are increasingly used for population-scale studies⁶, partly due to unified kit-based workflows. Several studies have compared these proteomic platforms with each other or MS-based approaches.

A recent study comparing three platforms (PEA based, aptamer based and MS (both discovery and targeted)) revealed limited protein overlap⁴². High-abundance protein depletion MS shared only 2,763 proteins with the SomaScan I1K Assay (targeting 9,685 proteins) and 1,664 with PEA methods (Explore HT, targeting 5,416 proteins)⁴². Median correlations between platform pairs averaged around 0.3, with even MS methods showing only a low median Spearman correlation of 0.34. High coefficients of variation (CVs), especially for low-abundance proteins,

likely account for these discrepancies⁴². Another study showed that over 60% of MS-detected proteins were quantified in both DDA and DIA modes (276 unique proteins)⁴³. However, overlap between MS and PEA was much smaller, with only 52 proteins in common⁴³. Only 35 proteins targeted by PEA assays were also detected by MS in both DDA and DIA methods, showing discrepancy and potential complementarity of the different technologies⁴³.

Several studies have compared PEA and SOMAms consistency^{44,45}. One study examined protein–genetic associations using both platforms (~6,000 proteins each), finding ~2,000 overlapping proteins with a median Spearman correlation of 0.33 (ref. 46). Another study compared protein quantitative trait loci from SomaScan I1K and Olink Explore HT using the plasma samples of 102 participants in the Atherosclerosis Risk in Communities Study, identifying 4,443 overlapping measurements with a median interplatform correlation of 0.14 (ref. 47). A third study measured 2,168 proteins in 3,976 Chinese adults using paired Olink Explore and SomaScan platforms, showing a median interplatform Spearman's ρ correlation of 0.29 (ref. 48).

The extreme dynamic concentration range of the circulation blood proteome creates distinct detection biases in proteomic analyses⁴². Owing to the wide dynamic concentration range of circulating proteins, MS-based proteomics has historically detected fewer low-abundance proteins, such as DNA polymerases, interferons or certain transcription factors. Conversely, proteins such as immunoglobulins and mitochondrial carrier proteins are often overrepresented in MS datasets. Affinity-based platforms, depending on the panel design, can sometimes mitigate this bias by selectively targeting proteins of lower abundance. Certain protein classes, such as complement components, calmodulin-related proteins and intracellular signaling molecules, are detected with both MS and affinity-based methods. These proteins are often more robustly detected and may represent a set of proteins that can be identified across platforms. However, while they may be detectable by both methods, the specificity of identification and the precision of quantification can vary, and measurements are still not always directly transferable between platforms.

These technology platforms measure different features of the circulating proteome, leading to complexities in comparing data from different assays that target different epitopes of a full-length protein or different proteoforms of a particular gene product. These discrepancies underscore the challenges of directly comparing results from separate studies and methodologies. Including a common reference in these assays would establish a collective understanding of core blood proteins, expand the range of detectable proteins, clarify discrepancies between existing and new technologies and refine proteoform-specific and method-dependent target identification.

Reference materials for cross-study circulating blood proteome

Building on the opportunities from the UK Biobank Pharma Proteomics Project⁴ and the challenges of large but fragmented proteomics data, studies of the circulating blood proteome clearly require standardization to ensure data comparability across studies and platforms, enabling comprehensive integration. All workflow components (Fig. 1) should be standardized, as proposed by the HUPO Human Plasma Proteome Project^{49,50}. Nevertheless, the enforcement of consistent protocols for the acquisition of all circulating blood proteome data is difficult due to the rapidly evolving landscape of analytically optimized MS-based workflows. Previous studies have attempted to develop standardized, end-to-end workflows for clinical specimens while emphasizing the inevitability of a diverse landscape of sample preparation techniques and LC–MS setups⁵¹.

Toward this end, HUPO community members have established a working group comprising the authors that integrates the different viewpoints presented here. Working group discussion yielded consensus that establishing standard reference materials provides a

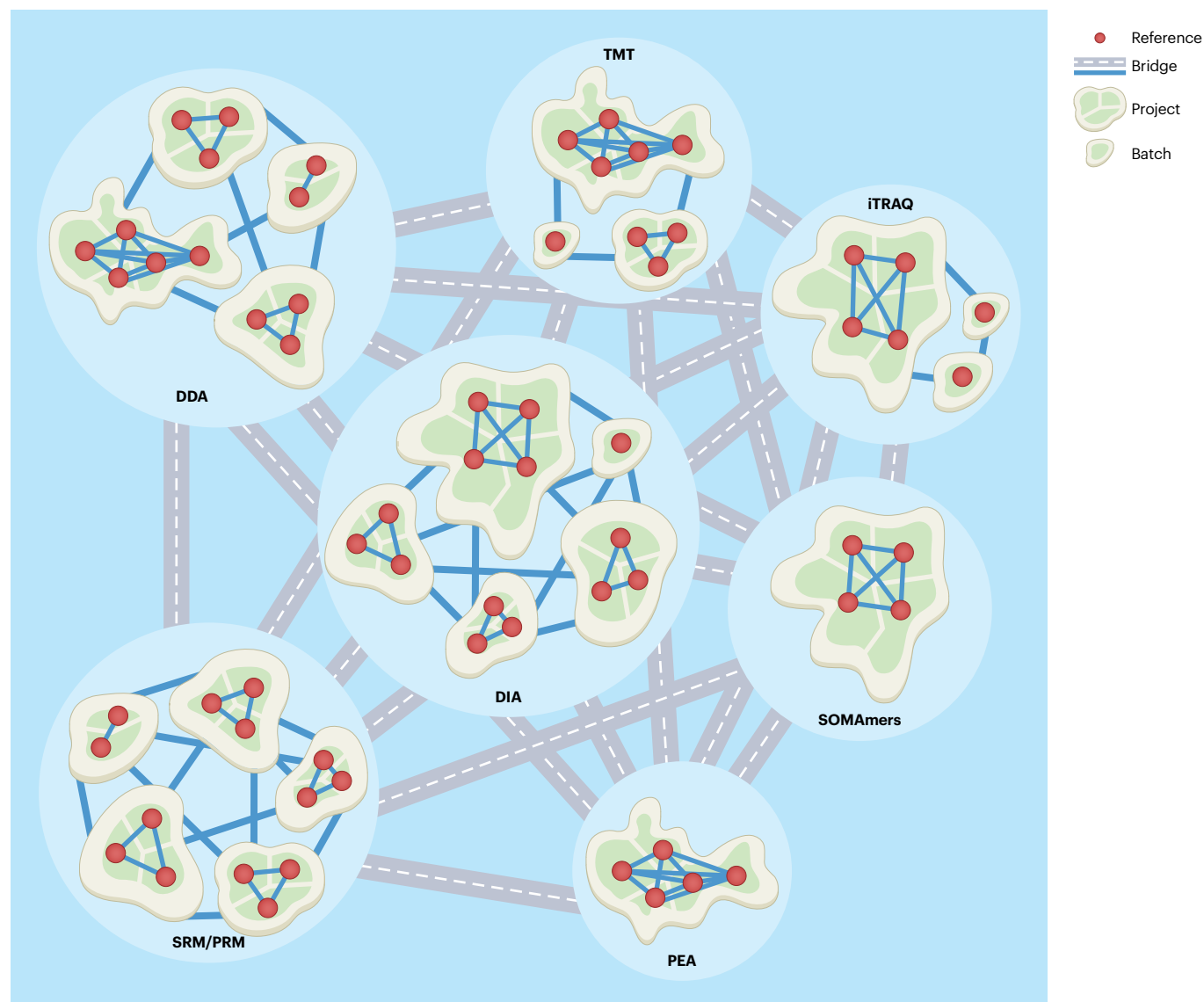


Fig. 2 | Potential of reference samples to allow comparability between diverse circulating blood proteome datasets. Data acquisition methods, such as DIA, DDA, tandem mass tagging (TMT), isobaric tags for relative and absolute quantitation (iTRAQ), selected reaction monitoring (SRM) and parallel reaction monitoring (PRM), PEA and SOMAmers, have generated vast amounts of isolated datasets (according to workflows depicted in Fig. 1), akin to islands in a sea,

separated by barriers. Introducing shared reference samples as proposed here (depicted as red points in the figure) can establish links not only within batches (regions of the same island) but also among different projects that apply the same method (different islands) and across distinct methodological approaches (different island groups).

more practical approach, serving as a universal benchmark with rigorously defined properties. Incorporating such standard references into experimental workflows allows researchers to normalize data across studies and platforms, ensuring comparability and enabling robust meta-analyses (Fig. 2). Data analysis strategies like the use of the sample-to-reference ratio (SRR) offer practical standardization and comparability approaches, demonstrated in multiomics Quartet project⁵² analyses of cell line-based DNA, RNA, protein and metabolite samples. Reference materials should be measured alongside study samples in identical analytical workflows, serving as the basis for data quality control and cross-study standardization. SRR serves as the quantitative result, ensuring protein-level comparability across platforms. SRR represents the ratio of quantitative measurements for identical proteins in both study samples and reference materials. SRR application enables strategies that minimize batch effects from factors, improve cross-platform data comparability and enhance proteomic research reproducibility.

We propose using two types of reference materials, donor-derived plasma and synthetic samples, and evaluate their respective advantages and limitations in Table 1. We note that the list of proteins will be curated continuously, because not all proteins are detectable at all sampling events and by all technologies. Protein levels may differ due to age-, sex- or disease-specific reasons⁴ and enter the circulation due to different processes like active secretion, shedding or tissue leakage^{53,54}. A key advantage of the donor-derived plasma reference samples is that their proteins behave similarly to those in study samples in terms of structural integrity, making them also suitable for affinity-based or other future assays. However, their practical utility may be limited by ethical and legal challenges related to the international shipment of plasma donations from human donors. These issues may be mitigated by commercial services that create pooled samples from multiple untraceable donors. Several standard reference materials for metabolomic research, such as SRM 1950 (refs. 55,56) and RM 8231 (ref. 57), have been commercialized and applied in metabolomic

Table 1 | Differences between types of reference material for circulation blood proteome studies

	Donor-derived reference materials	Synthetic reference materials
Biological relevance	Real samples but unable to accurately reflect the intricate spectrum of physiological and pathological states	Not real plasma but represents its core proteome
Cross-study data comparability	Requires interbatch calibration due to donor differences	Built-in consistency
Application scope	Compatible with MS and affinity-based tests	Compatible with MS; unclear for affinity-based tests
Quantification accuracy	Subject to variations between batches and temporal degradation	Built-in consistency
PTM detection capability	Retains native PTMs	Not applicable unless customized production
Cost and production difficulty	Relies on existing routines and cost models	Expensive to establish but cheap in production mode
Supply scalability	Relies on existing recruitment routines	Unlimited supply once established
Ethical issue	Consent required with the risk of potential withdrawal	No ethical issue

studies. However, owing to the exceptional complexity of proteomics, standardized reference materials for proteomic research have not yet been established. Furthermore, human-derived samples cannot be stored indefinitely, necessitating periodic calibration between successive production batches to establish conversion factors. This maintains access and continuity across different production lots and ensures an uninterrupted supply and use of reference materials.

Synthetic plasma samples can be created using synthesized peptides or engineered proteins representing core blood proteome(s). While more controllable than donor-derived plasma, synthetic samples may lack important nonprotein components. Whereas the design of peptide-based reference materials can largely follow published recommendations⁵⁸, the community will have to establish best practices for building protein-based reference materials to resemble the (pre-)analytical aspects of biological samples⁵⁹. These materials should use recombinant or (semi)synthetic proteins and be commercially available with regular updates and quality controls. Reliability requires controlled standard operating procedures for protein production, characterization, mixing, aliquoting and storage. Protein production systems should remain devoid of contaminants that could interfere with endogenous blood proteins. Protein characterization must assess purity, proteoform homogeneity, batch consistency and quantification accuracy. Protein mixing must replicate the complexity and dynamic range of the community-defined minimal circulation blood proteome. Different synthetic plasma versions can broaden reference ranges and study protein relationships. Finally, aliquoting and storage protocols should minimize freeze–thaw cycles while preserving protein integrity.

For core protein biomarkers, isotopically labeled peptides or proteins^{60–62} can be spiked into reference materials. These labeled reagents enable the transformation of relative quantification into absolute quantification. Unlike relative quantification, which relies on comparative measurements and requires normalization for methodological

variations, absolute quantification directly determines exact protein concentrations in blood. This method eliminates variability from sample preparation, instrument performance or matrix effects, enhancing reproducibility and reliability. The labeled peptides are structurally and chemically identical to endogenous counterparts, showing virtually no differences in retention time, ionization efficiency or fragmentation behavior. These characteristics make them ideal internal references, easily incorporated into samples during or after protease digestion. They also serve as reference analytes for developing methods targeting both labeled peptides and native equivalents. Heavy-to-light ratios (isotope-labeled-to-endogenous peptide signal ratios)⁶² provide a quantitative measure to correct for platform differences. Normalization strategies referencing endogenous peptides serve a similar purpose. Recently, common tryptic-digested pooled plasma and serum samples were used to assess MS-based proteomic workflows across eight laboratories, demonstrating reproducible detection and quantification and indicating high feasibility of using pooled digested reference samples⁶³. However, this approach may not be suited to evaluating differences between sample preparation methods, as many deplete full-length high-abundance proteins or enrich low-abundance proteins rather than their peptides. This strategy is also incompatible with affinity-based methods targeting binding epitopes in full-length proteins.

Compared to isotopically labeled peptides, isotopically labeled proteins better control digestion and sample preparation variability, enhancing quantification accuracy. This labeling also minimizes potential interferences in tandem mass spectra, improving fragment ion quantification reliability⁶⁴. Isotopically labeled proteins are typically produced through cell-free synthesis or cell-based metabolic labeling, both presenting challenges⁶⁵. While cell-free synthesis is costly, cell-based metabolic labeling suffers from incomplete labeling efficiency and label scrambling. However, a recent study achieved an average isotope incorporation of >99% for various ¹⁵N isotope-labeled reference proteins⁶². Integrating these protein references with targeted proteomic methods demonstrated the potential to rapidly and efficiently characterize biomarkers for alcohol-related liver disease. This approach showed high reproducibility and low CV (<10%), with adequate sensitivity and specificity, enabling swift translation of findings from cohort-based discovery research into clinical practice, for example, as MS-based clinical tests. For broad application and cross-study comparisons, generating diverse isotopically labeled protein references at competitive prices compared to those of isotope-labeled peptides is critical. Given extensive human proteoform diversity and current limitations in understanding their distinct functional roles across pathophysiological contexts, ensuring that isotopically labeled reference proteins perfectly replicate the conformational states and PTMs of native proteoforms is inherently challenging. Nevertheless, PTM-free isotopically labeled protein references can indirectly identify endogenous proteins with PTMs by identifying specific peptides with heavy-to-light ratios that deviate markedly from the average ratio of other peptides within the same protein, inferring the absence of unmodified endogenous peptides in the sample⁶².

Initial working plan

As discussed above, there are two main approaches for preparing reference materials: donor-derived plasma and synthetic plasma. Given the current inability to synthesize samples that adequately replicate critical proteoform variance and other biological components, we propose pooling donor-derived samples as the primary reference material in the first phase (Fig. 3a). Donors are selected based on strict health and demographic criteria. Standardized venipuncture protocols are used for the collection of plasma samples from fasting donors. EDTA is the preferred anticoagulant to maintain sample integrity. Samples are then systematically assembled to represent diverse geographic origins, ethnicities, age groups, genders and disease statuses. Sampling will

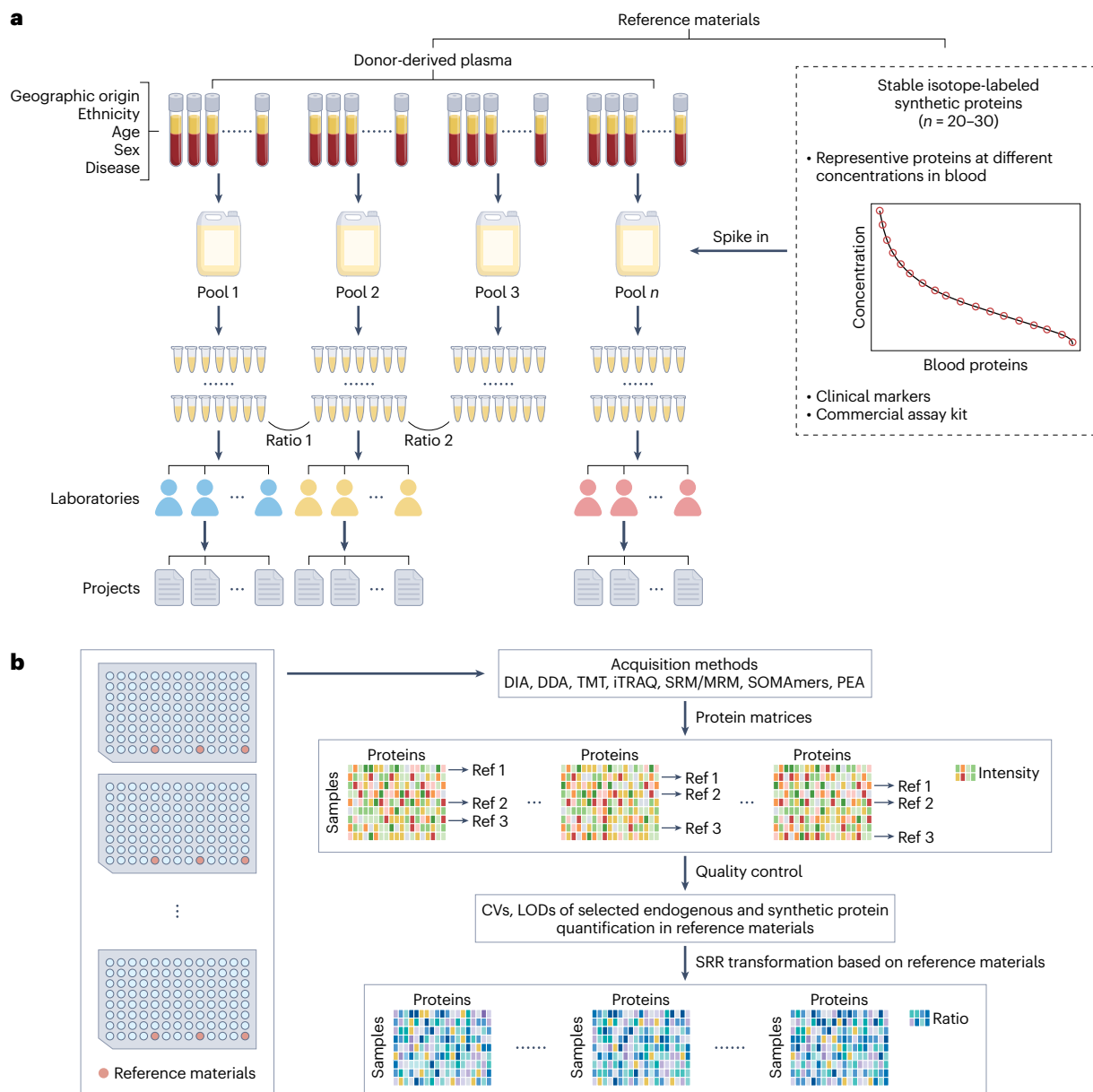


Fig. 3 | A preliminary standardization framework for reference materials in circulation blood proteomic studies. a, Workflow diagram illustrating the preparation of pooled reference materials. These references are derived from human plasma samples treated with DNase and/or RNase and systematically gathered from various countries. This collection aims to represent a wide range of ethnicities, ages, genders and health conditions. After mixing, the pooled plasma is combined with selected synthetic proteins and subsequently aliquoted for distribution to participating laboratories. The ratio between different reference materials refers to the quantitative ratio of proteins between different reference materials (such as those from different batches or different time

points). **b**, Reference material-based analytical pipeline, enabling reproducible cross-study comparisons. Each analysis plate includes three to four reference materials. Plates from different projects are processed using various acquisition methods, followed by protein identification and quantitative matrix generation. Quality control is then performed using the reference materials, evaluating the identification and quantification of endogenous proteins and synthetic proteins. Final SRR protein matrices are produced, facilitating data comparability across projects with diverse acquisition methodologies. LODs, limits of detection; MRM, multiple-reaction monitoring; ref, reference. Figure created using BioRender. lab, g. (2025) <https://BioRender.com/1w193ye>.

be conducted anonymously, with consent clearly indicating that this precludes withdrawal.

For standardized biological sample collection, two primary approaches are recommended to ensure quality and consistency. The first approach involves using certified reference repositories such as the Joint Research Centre, the National Institute of Metrology or the National Institute of Standards and Technology, which offer rigorously validated collection protocols. These institutions maintain the highest standards of quality control throughout the

collection process. Alternatively, organizations may opt for accredited third-party service providers specialized in biospecimen collection. These commercial entities offer scalable solutions while maintaining compliance with standardized protocols and operational efficiency, often providing additional logistical support. For studies requiring region-specific or disease-specific relevance, particularly those investigating ethnically diverse populations or rare proteoforms, researchers should obtain reference materials from certified local biobanks or vetted academic partners. These reference materials

BOX 1

Outstanding challenges for global references

- The composition of commonly used internal blood sample references varies between sources and studies due to characteristics of the donor, the protocols used to collect the samples and pre-analytical sample-handling effects⁵⁰.
- Data obtained from external reference samples may differ compared to local or technology-specific references and pools created from specific study samples.
- Protein profiles of identical sample types (for example, EDTA plasma) may differ between studies due to inclusion donor criteria, complicating cross-study comparisons.
- Further research is needed to harmonize sample collection procedures and apply these to the generation of reference material.
- Uncertainty about the ground truth composition and concentration of the expanding circulating proteome currently limits the community to establish universal reference values.
- Current proteomic methods provide single (relative) concentration values per protein rather than a comprehensive target analysis that includes the spectrum of molecular characteristics (for example, proteoforms).
- Protein isoforms have critical roles in the biology of the circulating proteome due to their diverse tissue-specific origins.
- Current reference samples will require continuous refinement and adaptation, as our understanding of the circulating proteome advances through technological improvements and collaborative community efforts.
- Differences in donor-derived reference samples will persist across future batches, necessitating strategies to mitigate this variability and enable cross-batch comparisons.
- The development, deployment and acceptance of universal reference materials, whether donor-derived, synthetic or hybrid, will continue to present significant challenges in the field.

serve as important complements to, rather than replacements for, universal standards.

While globally accessible sample pools with continuous updates (for example, every 3–5 years) would be optimal, logistical and ethical challenges related to cross-border sample transfer may arise. As an alternative, we propose creating multiple standardized plasma pools. We will select 20–30 core blood proteins based on factors outlined in Fig. 3a. Corresponding isotopically labeled versions at optimized concentrations that match endogenous levels will be incorporated to form the final reference materials. These isotopically labeled proteins will serve as internal references to assess reference material stability. The reference materials will then be aliquoted, labeled with expiration dates and distributed to proteomic laboratories worldwide. We suggest including three to four reference materials in every experimental batch (analysis plate) across all research projects for quality control, ensuring consistency in experimental procedures and data analysis (Fig. 3b). Reference material stability and assay performance, including intra-assay and interassay accuracy and precision, can be evaluated by calculating CVs and limits of detection for selected endogenous and synthetic proteins. The resulting proteome data, paired with these quality metrics, will enable robust cross-study comparisons in future research (Fig. 3b). Aliquoted samples from multiple plasma pools can

be analyzed simultaneously within an experimental batch to determine conversion ratios between pools. These ratios will enable quantitative conversion between reference material batches while maintaining longitudinal data comparability across updates (Fig. 3b).

The use of reference materials is expected to result in an approximate 5% increase in sample measurement costs, assuming that three to four samples per 96-well experiment are filled with reference materials. However, given the dramatic decrease in measurement costs associated with MS and affinity-based tests in recent years, the benefits of obtaining comparable data will outweigh this relatively modest cost increase.

This working plan represents our preliminary conceptual framework, which will undergo continuous refinement through practical implementation and collaborative discussions. These evolving updates may be formally documented in the future and include different types of plasma preparations (EDTA, heparin or citrate), serum samples, whole blood or even dried blood spot samples. We anticipate that there are multiple challenges for global references (Box 1).

Conclusion

Advanced studies of the circulating blood proteome hold great promise for precision medicine, but protocol variability across projects has created fragmented data that complicate data integration. While adopting a single protocol may not be feasible, we recommend including at least one shared reference plasma sample in future studies. Publicly depositing proteomic data from reference materials alongside study samples in controlled-access repositories is essential for rigorous cohort studies. Reference datasets should be treated as community resources to uphold data integrity. Journal editors might consider this a publication requirement and set limits on deviation between study and reference samples. Such references can bridge diverse datasets from different cohorts and phenotypes. Standardized reference samples are not merely technical refinements but a fundamental imperative for realizing the transformative potential of the circulating blood proteome. We further argue that limitations from varied platforms and methods could best be overcome by developing absolute rather than relative quantitative methods. By bridging current data fragmentation and fostering workflow interoperability, this initiative will pave the way for sustainable biomarker discovery, establishing common references that accelerate personalized medicine advancement. This can ultimately contribute to a more reliable, comprehensive and integrated understanding of human health and disease globally. Once sufficient comparable proteome data become available, we can develop advanced artificial intelligence models to better understand the blood proteome. If successful, the framework can be adapted to proteomic investigations of other biofluids facing comparable hurdles, including urine, cerebrospinal fluid, interstitial fluid and tear fluid, broadening protein-based biomarker identification in translational research.

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- Program, grant U24A20476); the National Key R&D Program of China (grants 2022YFF0608403 and 2021YFA1301600); the ‘Pioneer’ and ‘Leading Goose’ R&D Program of Zhejiang (grant 2024SSYS0035); NIH grants P30ES017885-11-S1 and U24CA271037 (G.S.O.), as well as the German Ministry of Education and Research (BMBF), as part of the National Research Node ‘Mass spectrometry in Systems Medicine (MSCoresys)’ under grant agreement 031L0220 to M.R. We thank π-Hub for supports to this work (grant number 2024-SP-008). Y.P.-R. thanks European Molecular Biology Laboratory core funding, Wellcome grants (208391/Z/17/Z and 223745/Z/21/Z) and the BBSRC grant ‘DIA-Exchange’ (BB/X001911/1). J.M.S. acknowledges the Knut and Alice Wallenberg Foundation for funding the Human Protein Atlas. We thank S. R. Piersma and T. V. Pham for their comments. We thank the HUPO community for advancing the field of proteomics.

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T.G., U.V., J.E.V.E. and J.M.S. conceived the idea and lead the discussion with all coauthors; X.C. and T.G. summarized the discussion and drafted the first draft. X.C., P.E.G., Y.P.-R., G.S.O., L.D., R.W., S.A., P.L., X.Y., C.C., M.R., C.R.J., Y. Zhao, Y.-J.C., T.C.W.P., N.B., L.S., X.D., Z.W., Y. Zhu, X.F., J.M.S., J.E.V.E., U.V. and T.G. revised the manuscript.

Competing interests

T.G. and Y. Zhu are shareholders of Westlake Omics. P.E.G. is an employee of Ions Biotechnologies. P.L. is an employee of Absea Biotechnology. N.B. is employee of Evosep. M.R. is the founder and shareholder of Eliptica. S.A. was a full-time employee of Alkahest, a Grifols company. J.M.S. is a scientific advisor for ABC Labs. The other authors do not have conflicts of interest related to this work.

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Peer review information *Nature Genetics* thanks Benjamin Sun and Christopher Whelan for their contribution to the peer review of this work.

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Acknowledgements

We are thankful for financial support from the following grants: the National Natural Science Foundation of China (Key Joint Research

¹Affiliated Hangzhou First People's Hospital, State Key Laboratory of Medical Proteomics, School of Medicine, Westlake University, Hangzhou, China. ²Westlake Laboratory of Life Sciences and Biomedicine, Westlake Center for Intelligent Proteomics, Hangzhou, China. ³Research Center for Industries of the Future, School of Life Sciences, Westlake University, Hangzhou, China. ⁴Ions.bio GmbH, Martinsried, Germany. ⁵European Molecular Biology Laboratory, European Bioinformatics Institute, Wellcome Trust Genome Campus, Hinxton, UK. ⁶Computational Medicine and Bioinformatics, University of Michigan, Ann Arbor, MI, USA. ⁷Internal Medicine, University of Michigan, Ann Arbor, MI, USA. ⁸Human Genetics, University of Michigan, Ann Arbor, MI, USA. ⁹Environmental Health, University of Michigan, Ann Arbor, MI, USA. ¹⁰Center for Advanced Measurement Science, National Institute of Metrology, Beijing, China. ¹¹Advanced Genomics Unit, Center for Research and Advanced Studies, Irapuato, Mexico. ¹²Alkahest, Inc., San Carlos, CA, USA. ¹³Absea Biotechnology GmbH, Berlin, Germany. ¹⁴State Key Laboratory of Medical Proteomics, Beijing Proteome Research Center, National Center for Protein Sciences—Beijing (PHOENIX Center), Beijing Institute of Lifeomics, Beijing, China. ¹⁵Department of Biochemistry, Charité – Universitätsmedizin Berlin, Berlin, Germany. ¹⁶OncoProteomics Laboratory, Department of Medical Oncology, Amsterdam UMC, location Vrije

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